INCH-POUND

MIL-F-44279A
21 September 1993
SUPERSEDING
MIL-F-44279(GL)
6 August 1987

MILITARY SPECIFICATION

FRUIT BARS, (OPERATIONAL RATION COMPONENT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fruit bars for use by the Department of Defense as a component of operational rations.

1.2 Classification. The fruit bars shall be of the following types as specified:

Type I - Fig
Type II - Blueberry

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8920

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SPECIFICATIONS

FEDERAL

A-A-1898	-	Cushioning Material, Cellulosic, Packaging
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

FEDERAL

FED-STD-595	-	Colors Used in Government Procurement

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

General Specification for Dairy Plants Approved for USDA Inspection and Grading Service (7 CFR Part 58)

(Copies are available from the Dairy Standardization Branch, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2750, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

U.S. Standards of Identity for Enriched Flour (21 CFR Part 137)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

FIG ADVISORY BOARD

Administrative Rules and Regulations for the Marketing Order for Dried Figs, As Amended

(Application for copies should be addressed to the Fig Advisory Board, P.O. Box 709, Fresno, CA 93712.)

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NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Blueberry filling. Blueberry filling shall be an all natural fruit filling containing natural blueberries and shall possess distinct blueberry odor and flavor. Other ingredients may include sugar, corn syrup, dextrose, modified corn starch, evaporated apples, natural flavors, blueberry juice concentrate, pectin, citric acid, salt, vegetable gums, and caramel color. Sodium benzoate may be used as a preservative. Fruit content shall be a minimum of 17 percent. Brix shall be 77 $\pm$ 2 percent and pH 3.4 $\pm$ 0.2 percent at 70°F (21°C). Brookfield viscosity shall be 40,000 centipoises $\pm$ 10,000 centipoises at 60°C at speed 20 while using spindle #7 (see 6.3.1).

3.2.2 Calcium phosphate, monobasic. Monobasic calcium phosphate shall comply with the requirements of the Food Chemicals Codex.

3.2.3 Citric acid, anhydrous, fine granulated. Fine granulated, anhydrous citric acid shall comply with the requirements of the Food Chemicals Codex.

3.2.4 Fig paste, dried, Adriatic and Mission fig blend. Adriatic and Mission blend fig paste shall be a ground blend of dried Adriatic figs (80 percent) and dried Mission figs (20 percent). The fruit shall comply with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder and also with the current "Administrative Rules and Regulations for the Marketing Order for Dried Figs". Figs shall be ground through a

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grinder with a plate that has inside diameter holes of 5/64 inch standard with 7/64 inch maximum. Product shall contain no additives and be prepared from the latest seasons crop (see 6.3.2).

3.2.5 Flour, general purpose, enriched. Flour shall be free-flowing, enriched, general purpose white flour and shall comply with U.S. Standard of Identity for Enriched Flour.

3.2.6 Salt. Salt shall be white refined sodium chloride with or without anticaking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.7 Shortening, vegetable. Vegetable shortening shall be refined, hydrogenated, deodorized vegetable oils and may contain antifoaming agents and antioxidants (active ingredients BHA, BHT, TBHQ, or propyl gallate) in amounts permitted by the FDA. The vegetable shortening shall possess a uniform plastic texture and shall have a stability of not less than 100 hours (A.O.M).

3.2.8 Sodium bicarbonate. Sodium bicarbonate shall comply with the requirements of the Food Chemicals Codex.

3.2.9 Sugar, white, granulated. Sugar shall be white refined, granulated cane or beet sugar, or a combination thereof.

3.2.10 Sugar, powdered or confectioner's. Powdered or confectioner's sugar shall be white and refined. The finished product shall contain not less than 96.5 percent sucrose, not more than 0.05 percent ash, not more than 0.80 percent moisture, and not less than 2.25 percent nor more than 3.25 percent of clean, edible starch thoroughly mixed, added to prevent caking. Not more than 2 percent shall remain on a US Standard No. 100 sieve and not less than 75 percent shall pass through a US Standard No. 200 sieve. The finished product shall not contain lumps larger than 1/2 inch in their greatest diameter that cannot be broken by light finger pressure.

3.2.11 Syrup, corn. Corn syrup shall be a clarified, concentrated, aqueous solution of the products obtained by partial hydrolysis of corn starch and includes dried corn syrup. The solids of corn syrup shall contain not less than 40 percent, by weight, of reducing sugars calculated as anhydrous dextrose.

3.2.12 Syrup, invert. Invert syrup shall be the aqueous solution of inverted or partially inverted sucrose. The syrup shall be colorless, odorless and flavorless except for sweetness. The syrup solids shall contain not more than 0.3 percent, by weight, of ash.

3.2.13 Whey protein concentrate, dry. The dry whey protein concentrate shall comply with the Food and Drug Administration regulations for Direct Food Substances Affirmed as Generally Recognized as Safe. The product shall be made from cheese whey which is pasteurized and spray dried. The whey

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time the chlorine is added to the water and the time when elapsing between the

protein concentrate shall be manufactured in a USDA approved plant that meets the requirements contained in the General Specification for Dairy Plants Approved for USDA Inspection and Grading Service. The dry whey protein concentrate shall be free flowing and not more than 60 days old from the time of manufacture. The product shall possess a pH value between 6.3 and 6.6 and a maximum moisture content of 5.0 percent.

3.2.14 Water. Water used for washing, formulation and processing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of fruit bar cookie jacket dough. The dough for the fruit bar cookie jacket shall be formulated and prepared as follows:

Ingredient	Percent by weight
Flour, general purpose, enriched	49.9
Sugar, powdered or confectioner's	14.8
Water	13.8
Shortening, vegetable	9.7
Syrup, corn	6.9
Syrup, invert	2.5
Whey protein concentrate, dry	1.2
Salt	0.5
Sodium bicarbonate	0.5
Calcium phosphate, monobasic	0.2

a. Powdered or confectioner's sugar, vegetable shortening, salt, and whey protein concentrate shall be mixed in a Hobart or other suitable mixer until mixture is a finely divided crumb or just creamed.

b. Invert syrup and corn syrup shall be added to the crumb and mixed well until smooth.

c. Flour, sodium bicarbonate, and monobasic calcium phosphate shall be mixed together until homogeneous.

d. The water and flour mixture shall be added gradually to the sugar, shortening mixture until a smooth dough is formed.

3.3.2 Preparation of fig bar filling. The fig bar filling shall be formulated and prepared as follows:

Ingredient	Percent by weight
Fig paste, dried, Adriatic and Mission fig blend	61.0

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Sugar, white, granulated	18.0
Water	8.8
Syrup, corn	6.0
Syrup, invert	6.0
Citric acid, anhydrous, fine granulated	0.1
Salt	0.1

a. Mix fig paste, granulated sugar, citric acid, and salt in a Hobart or other suitable mixer until homogeneous.

b. Add the corn syrup and invert syrup to the paste mixture and continue mixing until homogeneous.

c. Add water and continue mixing until homogeneous.

3.3.3 Preparation of blueberry bar filling. The blueberry bar filling shall comply with the requirements of 3.2.1.

3.3.4 Preparation of fruit bar. The fruit bar shall be mechanically formed so that the finished fruit bar consists of not less than 50 percent fruit bar filling. Water percent adjustments for dough and paste may be made for proper machine forming such that finished product requirements in 3.5 are achieved.

3.3.5 Baking. The bars shall be baked so that the dough is cooked and is a golden brown color.

3.4 Pouch filling and sealing. Each pouch shall contain two fruit bars and shall be filled and sealed such as to conform to the finished product requirements and to the requirements specified in 5.1.1.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty or moldy.

c. There shall be no color foreign to the product.

d. The contents of each pouch shall weigh not less that 54.0 grams.

e. Fruit bar cookie jacket shall be soft, but not soggy, and shall be a golden brown color.

f. Fig fruit bar filling shall be smooth except for the fig seeds and of an odor and flavor typical of high quality fig bars.

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g. Blueberry fruit bar filling shall be smooth except for the blueberry seeds and of an odor and flavor typical of blueberries.

h. Each bar shall measure not more than 0.7 inches thick, not more than 1.7 inches wide, and not more than 3.5 inches long.

i. The filling in the fruit bars shall be evenly distributed.

j. Fruit bars shall have cleanly and evenly cut ends and shall not be crushed or crumbled.

k. Fruit bars shall contain not less than 50 percent by weight of filling.

l. Moisture content shall be not greater than 18.0 percent.

3.5.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.6 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding of Human Food", and the plant sanitation requirements of the appropriate Government inspection agency.

3.7 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

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4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Pouch material certification. Material listed below may be accepted on the basis of a contractor's certification of conformance to the indicated requirements. In addition, compliance to the requirements for inside pouch dimensions and dimensions of manufacturer's seals may be verified by certificate of conformance.

Requirement	Requirement paragraph	Test procedure
Thickness of films for laminated material	5.1.1.1.1 and 5.1.1.2.1	As specified in L-P-378, except that a machinists' micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378
Aluminum foil thickness	5.1.1.1.1 and 5.1.1.2.1	As specified in QQ-A-1876

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laboratory has the responsibility for examining soil samples for parasitic ova NO

Laminated material identification and construction	5.1.1.1.1 and 5.1.1.2.1	Laboratory evaluation
Color of laminated material	5.1.1.1.1 and 5.1.1.2.1	Visual evaluation

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Results shall be reported to the nearest 0.1 gram. Any individual net weight of less than 54.0 grams shall be classified as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

4.5.4 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

TABLE I. Filled and sealed pouch defects 1/

Category		Defect
Major	Minor	
101		Tear, cut, hole, or abrasion through one or more layers in the pouch material or leakage through any heat seal 2/
102		Foldover wrinkle extending into the seal such that the effective closure seal is reduced to less than 1/16 inch 3/
103		Presence of entrapped product that reduces the effective closure seal to less than 1/16 inch 3/
104		Presence of delamination 4/
105		Unclean pouch 5/

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TABLE I. Filled and sealed pouch defects 1/ (cont'd)

Category		Defect
Major	Minor	
106		Labeling is missing, incorrect or illegible
107		Any impression or design on the heat seal surfaces which conceals or impairs visual detection of seal defects 6/
108		Less than 3/16 inch between inside edge of tear notch and inside edge of seal
109		Closure seal width not as specified
110		Closure seal not located as specified
111		Not packed as specified 7/
112		Presence of stress cracks in the aluminum foil 8/ 9/
	201	Tear notch or serrations missing or not located as specified
	202	Depth of tear notch or serrations not as specified
	203	Presence of delamination 4/

1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

2/ Pinholes or breaks inherent to the manufacturing process of the aluminum foil shall not be scored as a defect.

3/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide, from side seal to side seal that produces a hermetically sealed pouch.

4/ Delamination defect classification:

Major - Delamination of the outer ply in the pouch seal area that can be propagated to expose aluminum foil at the food product edge of the pouch after manual flexing of the delaminated area. To flex, the delaminated area shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall then be rapidly flexed 10 times by rotating both hands in

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alternating clockwise-counterclockwise directions. Care shall be exercised when flexing delaminated areas near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between thumb and forefinger and gently lifted toward the food product edge of the seal or if the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be classified as a major defect. Additionally, spot delamination of the outer ply in the body of the pouch that is able to be propagated beyond its initial borders is also a major defect. To determine if the laminated area is a defect, use the following procedure: Mark the outside edges of the delaminated area using a bold permanent marking pen. Open the pouch and remove the contents. Cut the pouch transversely not closer than 1/4 inch (± 1/16 inch) from the delaminated area. The pouch shall be flexed in the area in question using the procedure described above. Any propagation of the delaminated area, as evidenced by the delaminated area exceeding the limits of the outlined borders, shall be classified as a major defect.

Minor - Minor delamination of the outer ply in the pouch seal area is acceptable and shall not be classified as a minor defect unless it extends to within 1/16 inch of the food product edge of the seal. All other minor outer ply delamination in the pouch seal area or isolated spots of delamination in the body of the pouch that do not propagate when flexed as described above shall be classified as minor defects.

5/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings) or generally detracts from the clean appearance of the pouch. The following examples shall not be classified as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the pouch with a clean dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

6/ If doubt exists as to whether or not the sealing equipment leaves an impression or design on the closure seal surface that could conceal or impair visual detection of seal defects, samples shall be furnished to the contracting officer for a determination as to acceptability.

7/ Fruit bars shall be placed one on top of the other in a preformed pouch as specified in 5.1.1.1.2 or side by side in a form-fill-seal pouch as specified in 5.1.1.2.1.

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8/ Applicable to form-fill-seal pouches only.

9/ To test for stress cracks, the inside surface of both tray-shaped bodies shall be placed over a light source and the outside surface observed for the passage of light. Observation of light through the pouch material in the form of a curved or straight line greater than 2 mm in length shall be evidence of the presence of stress cracks in the aluminum foil.

4.5.5 Pouch leakage and seal separation testing. The filled and sealed pouches shall be tested by placing them in a dry desiccator, or similar apparatus, and subjecting them to a vacuum of 26 inches of mercury (atmospheric pressure is 29.9 inches of mercury) for 30 seconds. Any pouch that does not swell to a tightly distended form having at least one distorted edge while under the vacuum shall be classified as a major defect. Alternatively, pouches may be tested for leakage by submerging in water contained in a desiccator or other suitable container under a vacuum of 20 inches of mercury for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered as signs of leakage. Any pouch with one or more leaks shall be classified as a major defect. After leakage testing, the same pouches tested above shall be visually inspected for evidence of seal separation. Seal separation greater than 1/16 inch in the pouch manufacturer's seal shall be classified as a major defect. Any seal separation that reduces the effective closure seal width to less than 1/16 inch shall be classified as a major defect. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.6 Product examination. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE II. Product defects 1/ 2/

Category		Defect
Major	Minor	
101		Fruit bar cookie jacket hard or soggy; or not a golden brown color - (Underbaked: Soggy; Overbaked: Burnt, excessively dark and hard)
102		Fruit bar filling not smooth as specified
103		Fruit bar broken into more than three parts
	201	Flavor or odor not typical of high quality fig or blueberry fruit bar

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Material was dark golden-brown in color due

TABLE II. Product defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
	202	Size not as specified (see 3.5)
	203	Filling unevenly distributed 3/
	204	More than 1/4 of the fruit bar is crushed or crumbled

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, musty, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot (see 3.5.1).

3/ Unevenly distributed filling is filling that does not visually extend evenly throughout the interior of the jacket.

4.5.7 Examination for weight of filling. Eight filled and sealed pouches shall be randomly selected from the lot and the fruit bars shall be examined to determine conformance to the fruit "filling" weight requirement. The whole bars shall be weighed, the jackets removed, and the weight of the "filling" determined. The lot shall be rejected if the sample average "filling" weight is less than 50 percent of the total weight of the whole bars.

4.5.8 Moisture content testing. Eight filled and sealed pouches shall be selected at random from the lot regardless of lot size. The contents of the pouches shall be individually blended to uniformity using a blender or a food processor. The blending must be rapid and conducted in such a way that minimum heat is transferred to the product and that the product has minimum exposure to atmospheric moisture. The blended samples shall be tested for moisture content in accordance with the Official Methods of Analysis of the AOAC, method number 945.43. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the moisture content requirement in 3.5 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.9 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

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Major: Marking missing, incorrect, or illegible.
Cushioning material missing or not as specified.

Minor: More than 40 pounds packed in shipping container.
Pouches not packed flat in layers.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. Two fruit bars shall be unit packed in a barrier pouch as specified in 5.1.1.1 or 5.1.1.2.

5.1.1.1 Preformed pouches.

5.1.1.1.1 Pouch material. The preformed pouches shall be fabricated from 0.002 inch thick ionomer or polyethylene film laminated or extrusion coated to 0.00035 inch thick aluminum foil which is then laminated to 0.0005 inch thick polyester. The three plies shall be laminated with the polyester on the exterior of the pouch. All tolerances for thickness of pouch materials shall be plus or minus 20 percent. For Long Range Patrol (LRP) applications, the complete exterior surface of the pouch shall be uniformly colored in the range of green 34079 through 34087, or 24052 through 24087, or brown 30045 through 30118 (excluding 30109) or 10045 of FED-STD-595. For Ration Cold Weather (RCW) applications, the complete exterior surface of the pouch shall be colored white overall with a color in the range of 37778 through 37886 of FED-STD-595. The pouch material shall not impart any undesirable odor or flavor to the product and shall be FDA approved for food use. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches.

5.1.1.1.2 Pouch construction. The pouch shall be a flat style preformed pouch having inside dimensions of 3 inches wide by 5-7/8 inches long (± 1/8 inch in each dimension). The first dimension shall be the opening of the pouch between the heat sealed sides. The pouch shall be made by heat sealing three edges with 3/8 inch (minus 1/8 inch, plus 3/16 inch) wide seals. The heat seals shall be made in a manner that will assure hermetic seals. A "V", "C", or block "U" tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. A tear nick may be used in lieu of a tear notch. One side of the open end of the pouch may be provided with an extended or foldover lip, extended not more than 5/16 inch to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended or foldover lip. Alternatively, if the pouch has serrated edges, the serrations may be used as tear notches, provided that the serrations are sharp (no plastic tailings exists) and the serrations depth and the minimum seal width at the serrated

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edges are in accordance with the above requirements. The pouch shall be constructed with square or rounded corners at all four corners (radius approximately 3/8 inch).

5.1.1.1.3 Pouch filling and sealing. Two fruit bars shall be inserted into the pouch one on top of the other. The filled pouches shall be sealed under a vacuum level of eight to twelve inches of mercury with a minimum 1/8 inch wide heat seal. The closure seal shall not extend below the tear notch on either side of the pouch. The closure seal shall be free of foldover wrinkles or entrapped matter that reduces the effective closure seal width to less than 1/16 inch. The filled and sealed pouches shall withstand pouch leakage and seal separation testing as specified in 4.5.5.

5.1.1.2 Horizontal form-fill-seal pouches.

5.1.1.2.1 Pouch material. The horizontal form-fill-seal pouch shall consist of a formed tray-shaped body with a flat sheet, heat sealable cover or a tray-shaped body with a tray-shaped heat sealable cover. The tray-shaped body, the flat sheet cover and the tray-shaped cover shall be fabricated from a 3-ply flexible laminate barrier material consisting of, from outside to inside, 0.0009 inch thick oriented polypropylene bonded to 0.0007 inch thick deep forming aluminum foil with 10 pounds per ream pigmented polyethylene or adhesive bonded to 0.003 inch thick ionomer or a blend of not less than 50 percent linear low density polyethylene and polyethylene. The linear low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1 having a melt index range of 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range or 0.918 to 0.922 g/cc in accordance with ASTM D 1505. Alternatively, the tray-shaped body and flat-sheet cover combination package may be fabricated from material consisting of 0.0005 inch thick polyester laminated to 0.0007 inch thick deep forming aluminum foil that has 0.003 inch thick ionomer, or polyethylene film laminated or extrusion coated to the opposite side. The polyester shall be on the outside of the pouch. All tolerances for thickness of pouch materials shall be plus or minus 20 percent. The color requirements of the exterior (oriented polypropylene or polyester side) of the laminate shall be as specified in 5.1.1.1.1. The pouches shall not impart any odor or flavor to the product and shall be FDA approved for food use.

5.1.1.2.2 Pouch construction. The tray-shaped body and the tray-shaped cover shall be formed by drawing the flexible laminate material into an appropriately shaped cavity. The flat cover shall be in the form of a flat-sheet of the barrier material taken from roll stock. Two fruit bars shall be placed side by side in the tray shaped body of the pouch. The filled pouch body shall be hermetically sealed with a vacuum level of 15-17 inches of mercury when horizontal form-fill-seal equipment is used. Pouch closure shall be effected by heat sealing together the cover and body along the entire pouch perimeter. The outside dimensions of the sealed pouches shall be 5-1/2 inches wide by 5-1/2 inches long (± 1/8 inch in each

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dimension). The closure seal width shall be 3/8 inch (plus 3/16 inch, minus 1/8 inch). A tear nick or a tear notch shall be placed on one outside edge or two opposite outside edges of the pouch. Each tear nick or tear notch shall be located not more than 1-1/4 inches or less than 1/4 inch from an outside corner of the pouch. Alternatively, if the pouch has serrated edges, the serrations may be used as tear notches provided that the serrations are located to effect easy opening in the machine direction of the pouch laminates; the serrations are sharp (no plastic tailings exist); and the serration depth and minimum seal width at the serrations are specified for the tear nick or notch. The sealed pouches shall not show any evidence of material degradation, aluminum stress cracking, delamination or foreign odor. Heat seals shall be free of foldover wrinkles and occluded matter. The filled and sealed pouches shall withstand pouch leakage and seal separation testing as specified in 4.5.5.

5.2 Packing. Packing shall be level C.

5.2.1 Level C packing. Not more than 40 pounds of pouched product shall be packed flat in layers in a fiberboard box constructed and closed in accordance with style RSC-L, class domestic, variety SW, grade 200 of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to grade II, class B or C of A-A-1898 or cushioning material not less than 1/16 inch thick, conforming to type I of PPP-C-1797 or cushioning material conforming to type VII, class 1 of PPP-C-1752 shall be placed between each of the layers and in the bottom and top of the box to minimize movement of the individual packages.

5.3 Labeling and marking.

5.3.1 Pouches. Each pouch shall be clearly printed or stamped, in a manner that does not damage the pouch, with permanent black ink which is free of carcinogenic elements or ingredients. The information shall be located on the body of the pouch not closer than 1/16 inch to any seal. If a non-contact type printer is used, the information may be located anywhere on the pouch (in one complete print), except the closure seal area. The label shall contain the following information:

Product name (letters not less than 7/32 inch high)
Date 1/ (letters not less than 1/8 inch high)
Net weight (letters not less than 1/8 inch high)
Contractor's name and address (letters not less than 3/32 inch high)

1/ Each pouch shall have the date of pack noted by using a four digit code beginning with the final digit of the current year followed by the three digit Julian day code. For example, March 19, 1991 would read 1079. The Julian day code shall represent the date the product is unit packed into the pouch.

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Alternatively, the fruit bars in original vendor packaging may be packaged in a barrier pouch. If ingredient markings are missing, they shall be added to the barrier pouch. Nutrition labeling shall be in accordance with the provisions of 21 CFR Part 101.9, Nutrition labeling of foods.

5.3.2 Shipping container. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of product required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.5.1 and 6.2).
f. When fruit bar is procured as a component of the LRP or RCW.

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instruction in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Ingredient information.

6.3.1 Blueberry filling. It has been found that Natural Blueberry Filling WL-11785 manufactured by Florasynth, Inc, meets the requirements of 3.2.1 and performs satisfactorily in this product.

6.3.2 Fig paste, dried, Adriatic and Mission fig blend. It has been found that Blue Ribbon Fig Paste #00619 80:20 Adriatic:Mission Figs manufactured by Sun Diamond Growers, meets the requirements of 3.2.4 and performs satisfactorily in this product.

6.4 Subject term (key word) listing.

Blueberry bars
Cookie bars
Fig bars

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Long Range Patrol
Pouched food
Ration Cold Weather
Shelf stable foods

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

Preparing activity:

Army - GL

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-44279A	1993 September 21

3. DOCUMENT TITLE

FRUIT BARS, (OPERATIONAL RATION COMPONENT)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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